

lab04_leila_kawanishi_277185

Leila Kawanishi

Laboratório 4: Importação de Dados Retangulares - Excel

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Conjunto de Dados

O conjunto de dados em questão é o conjunto de índices de desenvolvimento mundial, disponibilizado pelo Banco Mundial através do link <http://datatopics.worldbank.org/world-development-indicators/>.

Expectativas

Ao fim deste laboratório, você deverá ser capaz de:

- Importar dados de planilhas específicas de arquivos em formato Excel;
- Trabalhar com variáveis cujos nomes fogem do padrão usual de programação (por exemplo, variáveis com espaços brancos em seus nomes);
- Selecionar variáveis do interesse;
- Verificar se os dados estão no formato tidy e transformá-los para tal formato;
- Optar apropriadamente entre filtering joins e mutating joins;
- Produzir visualizações de dados de séries temporais;

Atividades

Q1. Carregue os pacotes readxl e tidyverse.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readxl)
```

Q2. Crie uma variável chamada `fname` que contenha o caminho para o arquivo `WDIEXCEL.xlsx`, disponível no seu diretório de dados. Confirme que o R identifica a existência do arquivo utilizando o comando `file.exists`

```
fname <- "../WDIEXCEL.xlsx"
file.exists(fname)
```

```
[1] TRUE
```

```
excel_sheets(fname)
```

```
[1] "Data"          "Country"       "Series"        "country-series"
[5] "series-time"   "footnote"
```

Q3. Leia a planilha “Country” e armazene no objeto `countries`.

```
countries <- read_excel(fname, sheet = "Country")
countries
```

```
# A tibble: 264 x 31
  `Country Code` `Short Name`      `Table Name` `Long Name` `2-alpha code`
  <chr>          <chr>          <chr>        <chr>      <chr>
1 ABW           Aruba           Aruba        Aruba      AW
2 AFE           Africa Eastern and So~ Africa East~ Africa Eas~ ZH
3 AFG           Afghanistan    Afghanistan  Islamic St~ AF
4 AFW           Africa Western and Ce~ Africa West~ Africa Wes~ ZI
```

```

5 AGO          Angola          Angola      People's R~ AO
6 ALB          Albania         Albania     Republic o~ AL
7 AND          Andorra         Andorra     Principali~ AD
8 ARB          Arab World      Arab World  Arab World  1A
9 ARE          United Arab Emirates United Arab~ United Ara~ AE
10 ARG         Argentina       Argentina   Argentine ~ AR
# i 254 more rows
# i 26 more variables: `Currency Unit` <chr>, `Special Notes` <chr>,
#   Region <chr>, `Income Group` <chr>, `WB-2 code` <chr>,
#   `National accounts base year` <chr>,
#   `National accounts reference year` <dbl>, `SNA price valuation` <chr>,
#   `Lending category` <chr>, `Other groups` <chr>,
#   `System of National Accounts` <chr>, ...

```

Q4. Mantenha apenas os registros de 5 países (Brasil e outros 4 de sua escolha).

```

countries <- countries %>%
  filter(`Short Name` %in% c("Brazil", "Nigeria", "Georgia", "Japan", "Fiji")) %>%
  select(`Country Code`, `Short Name`)

```

Q5. Leia a planilha “Data” e armazene no objeto WDI.

```

WDI <- read_excel(fname, sheet = "Data")
head(WDI)

```

```

# A tibble: 6 x 70
  `Country Name` `Country Code` `Indicator Name` `Indicator Code` `1960` `1961`
  <chr>          <chr>          <chr>          <chr>          <dbl> <dbl>
1 Africa Eastern~ AFE          Access to clean~ EG.CFT.ACCS.ZS      NA      NA
2 Africa Eastern~ AFE          Access to clean~ EG.CFT.ACCS.RU.~    NA      NA
3 Africa Eastern~ AFE          Access to clean~ EG.CFT.ACCS.UR.~    NA      NA
4 Africa Eastern~ AFE          Access to elect~ EG.ELC.ACCS.ZS      NA      NA
5 Africa Eastern~ AFE          Access to elect~ EG.ELC.ACCS.RU.~    NA      NA
6 Africa Eastern~ AFE          Access to elect~ EG.ELC.ACCS.UR.~    NA      NA
# i 64 more variables: `1962` <dbl>, `1963` <dbl>, `1964` <dbl>, `1965` <dbl>,
#   `1966` <dbl>, `1967` <dbl>, `1968` <dbl>, `1969` <dbl>, `1970` <dbl>,
#   `1971` <dbl>, `1972` <dbl>, `1973` <dbl>, `1974` <dbl>, `1975` <dbl>,
#   `1976` <dbl>, `1977` <dbl>, `1978` <dbl>, `1979` <dbl>, `1980` <dbl>,
#   `1981` <dbl>, `1982` <dbl>, `1983` <dbl>, `1984` <dbl>, `1985` <dbl>,
#   `1986` <dbl>, `1987` <dbl>, `1988` <dbl>, `1989` <dbl>, `1990` <dbl>,
#   `1991` <dbl>, `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, `1995` <dbl>, ...

```

Q6. Seleccione apenas as colunas:

- Country Code
- Indicator Code
- E todos os anos de 1960-2018

```
WDI <- WDI %>%
  select(`Country Code`, `Indicator Code`, `1960`:`2018`
)
glimpse(WDI)
```

Rows: 396,970

Columns: 61

```
$ `Country Code`      <chr> "AFE", "AFE", "AFE", "AFE", "AFE", "AFE", "AFE", "AFE~
$ `Indicator Code`    <chr> "EG.CFT.ACCS.ZS", "EG.CFT.ACCS.RU.ZS", "EG.CFT.ACCS.U~
$ `1960`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1961`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1962`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1963`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1964`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1965`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1966`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1967`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1968`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1969`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1970`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1971`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1972`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1973`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1974`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1975`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1976`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1977`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1978`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1979`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1980`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1981`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1982`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1983`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1984`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1985`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1986`              <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
```

```

$ `1987`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1988`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1989`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1990`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1991`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1992`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1993`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1994`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1995`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1996`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1997`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1998`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `1999`      <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~
$ `2000`      <dbl> 1.149470e+01, 3.548167e+00, 3.235069e+01, 1.991726e+0~
$ `2001`      <dbl> 1.184106e+01, 3.704024e+00, 3.268334e+01, 1.994428e+0~
$ `2002`      <dbl> 1.220252e+01, 3.890043e+00, 3.310018e+01, 2.154865e+0~
$ `2003`      <dbl> 1.257338e+01, 4.091720e+00, 3.349046e+01, 2.249322e+0~
$ `2004`      <dbl> 1.295869e+01, 4.274249e+00, 3.399118e+01, 2.372445e+0~
$ `2005`      <dbl> 1.337114e+01, 4.532521e+00, 3.447061e+01, 2.344954e+0~
$ `2006`      <dbl> 1.378271e+01, 4.757490e+00, 3.492266e+01, 2.517248e+0~
$ `2007`      <dbl> 1.425418e+01, 5.004035e+00, 3.550103e+01, 2.679514e+0~
$ `2008`      <dbl> 1.473659e+01, 5.248924e+00, 3.591535e+01, 2.590885e+0~
$ `2009`      <dbl> 1.517156e+01, 5.499747e+00, 3.643522e+01, 2.618785e+0~
$ `2010`      <dbl> 1.562017e+01, 5.747851e+00, 3.682269e+01, 2.741495e+0~
$ `2011`      <dbl> 1.612320e+01, 6.028196e+00, 3.725257e+01, 2.890976e+0~
$ `2012`      <dbl> 1.659973e+01, 6.309218e+00, 3.763599e+01, 3.167343e+0~
$ `2013`      <dbl> 1.706105e+01, 6.643625e+00, 3.796932e+01, 3.172357e+0~
$ `2014`      <dbl> 1.763415e+01, 6.932310e+00, 3.834023e+01, 3.187196e+0~
$ `2015`      <dbl> 1.814583e+01, 7.259936e+00, 3.874199e+01, 3.392228e+0~
$ `2016`      <dbl> 1.868512e+01, 7.606712e+00, 3.905263e+01, 3.885960e+0~
$ `2017`      <dbl> 1.920563e+01, 7.926604e+00, 3.932107e+01, 4.022374e+0~
$ `2018`      <dbl> 1.974277e+01, 8.309896e+00, 3.964953e+01, 4.303507e+0~

```

Q7. Confirme se os dados wm WDI estão em formato tidy. Se não estiverem, transforme-os para tal formato. Faça de forma que exista uma coluna referente ao ano chamada YEAR.

#WDI nao está em formato tidy porque a variavel ano esta distribuída em colunas. Vamos usar o

```

WDI <- WDI %>%
  pivot_longer(
    cols = `1960`: `2018`,
    names_to = "YEAR",

```

```

      values_to = "value"
    )
  glimpse(WDI)

```

Rows: 23,421,230

Columns: 4

```

$ `Country Code`  <chr> "AFE", "AFE", "AFE", "AFE", "AFE", "AFE", "AFE", "AFE~
$ `Indicator Code` <chr> "EG.CFT.ACCS.ZS", "EG.CFT.ACCS.ZS", "EG.CFT.ACCS.ZS", ~
$ YEAR            <chr> "1960", "1961", "1962", "1963", "1964", "1965", "1966~
$ value           <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N~

```

Q8. Confirme se a coluna referente ao ano está em formato inteiro.

#Como no glimpse acima, "YEAR" esta em formato character. Vamos transforma em inteiro

```

WDI <- WDI %>%
  mutate(YEAR = as.integer(YEAR))
class(WDI$YEAR)

```

```
[1] "integer"
```

Q9. Para o objeto WDI, mantenha apenas os registros dos países listados em countries. Selecione as colunas:

- Country Code
- Short Name
- YEAR
- SP.DYN.CBRT.IN

```

WDI <- WDI %>%
  inner_join(
    countries %>% select(`Country Code`, `Short Name`),
    by = "Country Code"
  ) %>%
  filter(`Indicator Code` == "SP.DYN.CBRT.IN") %>%
  select(
    `Country Code`,
    `Short Name`,
    YEAR,
    value
  )

```

```

) %>%
  rename(`SP.DYN.CBRT.IN` = value)

glimpse(WDI)

```

```

Rows: 295
Columns: 4
$ `Country Code` <chr> "BRA", "BRA", "BRA", "BRA", "BRA", "BRA", "BRA", "BRA",~
$ `Short Name`    <chr> "Brazil", "Brazil", "Brazil", "Brazil", "Brazil", "Brazil", "Braz~
$ YEAR            <int> 1960, 1961, 1962, 1963, 1964, 1965, 1966, 1967, 1968, 1~
$ SP.DYN.CBRT.IN <dbl> 43.850, 43.292, 42.698, 42.014, 41.001, 39.844, 38.638,~

```

Q10. Ordene o objeto de forma que, para cada país, as informações estejam ordenadas por ano.

```

WDI <- WDI %>%
  arrange(`Short Name`, YEAR)

glimpse(WDI)

```

```

Rows: 295
Columns: 4
$ `Country Code` <chr> "BRA", "BRA", "BRA", "BRA", "BRA", "BRA", "BRA", "BRA",~
$ `Short Name`    <chr> "Brazil", "Brazil", "Brazil", "Brazil", "Brazil", "Brazil", "Braz~
$ YEAR            <int> 1960, 1961, 1962, 1963, 1964, 1965, 1966, 1967, 1968, 1~
$ SP.DYN.CBRT.IN <dbl> 43.850, 43.292, 42.698, 42.014, 41.001, 39.844, 38.638,~

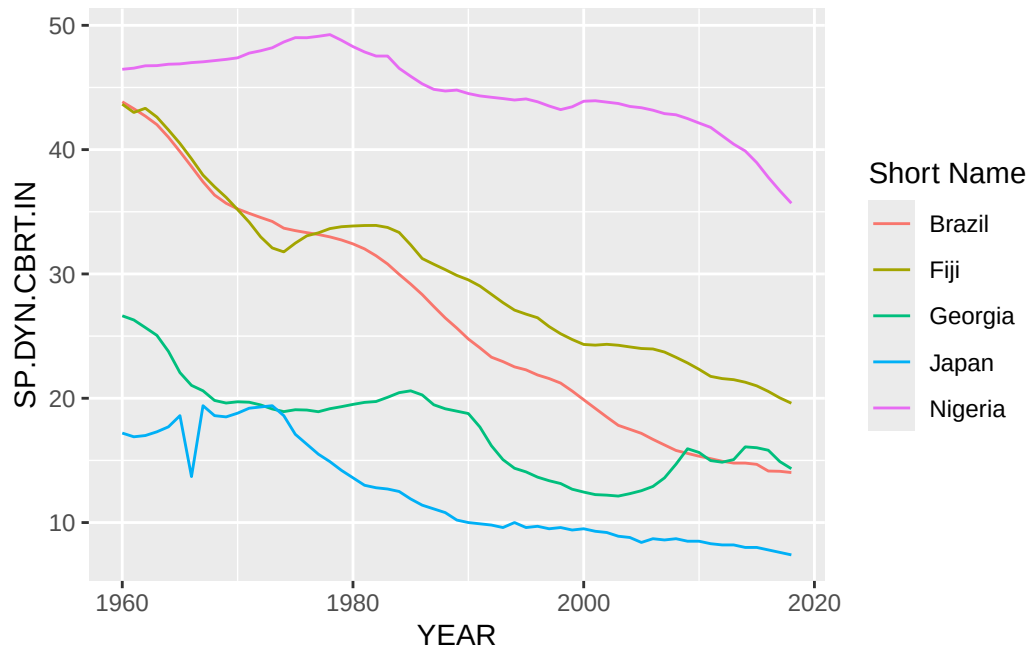
```

Q11. Construa um gráfico de linhas, usando o pacote ggplot2, que tenha no eixo X a variável YEAR, no eixo Y a variável SP.DYN.CBRT.IN e que cada linha corresponda a um país diferente.

```

ggplot(WDI, aes(x = YEAR,
                y = `SP.DYN.CBRT.IN`,
                color = `Short Name`)) +
  geom_line()

```



```
Sys.setenv(TZ = "America/Sao_Paulo")

paste(
  "Arquivo compilado em:",
  format(Sys.time(), "%d/%m/%Y às %H:%M:%S")
)
```

```
[1] "Arquivo compilado em: 08/09/2026 às 08:04:14"
```